



NUS | Computing
National University
of Singapore

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Computer Science

T07 – Week 8

Pre-Midterm Exam Discussion

CS3230 – T11

$n^{10} - n^9$ is in

- A** $\Omega(n^{11})$
- B** $o(n^{10})$
- C** $\Theta(n^9)$
- D** $O(n^8)$
- E** None of the above

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Solution

Since $\lim_{n \rightarrow \infty} \frac{n^{10} - n^9}{n^{10}} = \lim_{n \rightarrow \infty} \left(1 - \frac{1}{n}\right) = 1 \implies n^{10} - n^9 \in \Theta(n^{10})$, the correct answer is None of the above.

$(n + 1)!$ is in

- A** $O(n!)$
- B** $\omega(n!)$
- C** $\Theta(n!)$
- D** $o(n!)$
- E** None of the above

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Solution

$$\lim_{n \rightarrow \infty} \frac{(n+1)!}{n!} = \lim_{n \rightarrow \infty} (n+1) = \infty \implies (n+1)! \in \omega(n!).$$

$2^{\log_3 n}$ is in

- ☐ A $O(\log_2 n)$
- ☐ B $\Theta(n^2)$
- ☐ C $\omega(n)$
- ☐ D $\Omega(\sqrt{n})$
- ☐ E None of the above

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Solution

$2^{\log_3 n} = n^{\log_3 2} = n^{0.6309\dots}$, so options A, B, and C are incorrect. We check for D:
 $\lim_{n \rightarrow \infty} \frac{2^{\log_3 n}}{\sqrt{n}} = \lim_{n \rightarrow \infty} \frac{n^{\log_3 2}}{n^{1/2}} = \lim_{n \rightarrow \infty} n^{\log_3 2 - 1/2} = \infty \implies 2^{\log_3 n} \in \Omega(\sqrt{n})$.

Suppose $f(n) \in \Theta(n^2(\log n)^5)$ and $g(n) \in \Theta(n^5(\log n)^2)$. Then, $f(n) + g(n)$ is in

- A** $\Theta(n^5(\log n)^5)$
- B** $\Theta(n^2(\log n)^5)$
- C** $\Theta(n^5(\log n)^2)$
- D** $\Theta(n^7(\log n)^7)$
- E** None of the above

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- E** None of the above

Solution

$$\lim_{n \rightarrow \infty} \frac{f(n)}{g(n)} = \lim_{n \rightarrow \infty} \frac{n^2(\log n)^5}{n^5(\log n)^2} = \lim_{n \rightarrow \infty} \frac{(\log n)^3}{n^3} = 0 \implies f(n) \in o(g(n)). \text{ Hence,}$$
$$\lim_{n \rightarrow \infty} \frac{f(n)+g(n)}{g(n)} = \lim_{n \rightarrow \infty} \frac{f(n)}{g(n)} + 1 = 0 + 1 = 1 \implies f(n) + g(n) \in \Theta(n^5(\log n)^2).$$

Suppose $T(n) = 36T(n/6) + 2n + n^{8/3}$. Then, $T(n)$ is in

- A** $\Theta(n^{8/3})$
- B** $\Theta(n^{8/3} \log n)$
- C** $\Theta(n^2)$
- D** $\Theta(n^2 \log n)$
- E** None of the above

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- E** None of the above

Solution

Since $a = 36$, $b = 6$, $d = \log_6 36 = 2$, and $f(n) = 2n + n^{8/3} \in \Omega(n^{2+\epsilon})$ with $\epsilon = \frac{8}{3} - 2 = \frac{2}{3}$, and the regularity condition holds (e.g., $36 \cdot f(n/6) \leq \frac{1}{6^{1/3}} f(n)$ for large n , with $\frac{1}{6^{1/3}} < 1$), by Master Theorem Case 3 we have $T(n) \in \Theta(n^{8/3})$.

Suppose $T(n) = 64T(n/4) + 3n^{1.5}$. Then, $T(n)$ is in

- A** $\Theta(n^2)$
- B** $\Theta(n^3)$
- C** $\Theta(n^{1.5})$
- D** $\Theta(n^{1.5} \log n)$
- E** None of the above

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- E** None of the above

Solution

Since $a = 64$, $b = 4$, $d = \log_4 64 = 3$, and $f(n) = 3n^{1.5} \in O(n^{3-\epsilon})$ with $\epsilon = 1.5$, by Master Theorem Case 1 we have $T(n) \in \Theta(n^3)$.

Suppose $T(n) = T(n/5) + 2T(n/3) + n$. Then, $T(n)$ is in

- A** $\Theta(n)$
- B** $\omega(n^2)$
- C** $\Omega(n \log n)$
- D** $o(n)$
- E** None of the above

Suppose $T(n) = T(n/5) + 2T(n/3) + n$. Then, $T(n)$ is in

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- E** None of the above

Solution

Clearly, $T(n) \geq n$. Let $c \geq \frac{15}{2}$ be such that $T(n) \leq cn$ for all $n \leq 100$. We will show by induction that $T(n) \leq cn$ for all n . Assuming that this is true for all $n < n_0$ where $n_0 > 100$, we have $T(n_0) \leq c \cdot \frac{n_0}{5} + 2c \cdot \frac{n_0}{3} + n_0 \leq cn_0$, where the last inequality follows from the assumption that $c \geq \frac{15}{2}$. Hence, $T(n) \in \Theta(n)$.

For any randomized algorithm, let $E(n)$ and $T(n)$ denote the expected and worst-case running time, respectively, for inputs of length n . Then, which of the following statement is always **TRUE**, irrespective of the randomized algorithm being considered?

- A** For every n , $E(n) < T(n)$
- B** For every n , $E(n) = T(n)$
- C** For every n , $E(n) > T(n)$
- D** For at least one n , $E(n) < T(n)$, and for at least one n , $E(n) > T(n)$
- E** None of the above

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- D** For at least one n , $E(n) < T(n)$, and for at least one n , $E(n) > T(n)$
- E** None of the above

Solution

Since $T(n)$ is the maximum running time over all inputs and random choices, we always have $E(n) \leq T(n)$. However, it can happen that $E(n) < T(n)$ for some n (possibly none), and $E(n) = T(n)$ for the remaining n (possibly none).

Suppose we throw 3 balls independently and uniformly at random into 5 bins. Then,

- A** The probability that all the balls fall into the same bin is 0.
- B** The probability that all the balls fall into the same bin is $\frac{3}{5}$.
- C** The probability that all the balls fall into the same bin is $\frac{1}{25}$.
- D** The probability that all the balls fall into the same bin is $\frac{1}{9}$.
- E** None of the above.

Suppose we throw 3 balls independently and uniformly at random into 5 bins. Then,

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- D** The probability that all the balls fall into the same bin is $\frac{1}{9}$.
- E** None of the above.

Solution

Let B_i be the event that all 3 balls fall in bin i . Then, $\Pr(B_i) = (1/5)^3 = 1/125$, and since the events $B_1, \dots, B_5$ are disjoint (never occur at the same time),
 $\Pr(\bigcup_{i=1}^5 B_i) = 5 \cdot (1/125) = 1/25$.

Question A.10

Consider an undirected graph $G = (V, E)$ with $n = |V|$ vertices and $m = |E|$ edges. A randomized algorithm selects a vertex $v \in V$ uniformly at random and returns $\deg(v)$, where $\deg(v)$ denotes the degree of vertex v . Let X be the random variable that denotes the output of this algorithm. What is the expected value of X , i.e., $\mathbb{E}[X]$?

- A** m
- B** n
- C** m/n
- D** $2m/n$
- E** None of the above

Question A.10

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- ☐ A m
- ☐ B n
- ☐ C m/n
- ☐ D $2m/n$
- ☐ E None of the above

Solution

From the [handshaking lemma](#), we have $\sum_{u \in V} \deg(u) = 2m$. Since the vertex is chosen uniformly at random from V , the expected value is $\mathbb{E}[X] = \frac{1}{n} \sum_{v \in V} \deg(v) = \frac{2m}{n}$.

Question B.1

There are three rods and $n \geq 1$ disks of different diameters, all stacked on the first rod, smallest on top and largest at the bottom.

Alice must move all disks to the third rod, following these rules:

- 1 Move one disk at a time from the top of any rod.
- 2 No disk may be placed on a smaller disk.
- 3 (Variant) Disks can only move between adjacent rods.

Let $f(n)$ be the number of moves Alice needs to complete this task (minimize moves, but no proof required).

- a. Write a recurrence for $f(n)$, including the base case(s), and explain how you derived it. $\square$
- b. Solve the recurrence from part (a) (i.e., give a closed-form formula for $f(n)$, with justification). $\square$

Solution a

Alice can make the following moves:

Step	Move	From	To	Moves
1	Top $n - 1$ disks	First rod	Third rod	$f(n - 1)$
2	Bottom disk	First rod	Second rod	1
3	Top $n - 1$ disks	Third rod	First rod	$f(n - 1)$
4	Bottom disk	Second rod	Third rod	1
5	Top $n - 1$ disks	First rod	Third rod	$f(n - 1)$

Hence,

- Base case: $f(1) = 2$ moves (first rod to the second, second rod to the third)
- Recursive step: $3f(n - 1) + 2$ moves

$$f(n) = \begin{cases} 2, & \text{if } n = 1, \\ 3f(n - 1) + 2, & \text{if } n > 1. \end{cases}$$

Solution b

We claim that $f(n) = 3^n - 1$ and prove this by induction.

Base case

$$f(1) = 2 = 3^1 - 1.$$

Inductive step

Assume the claim holds for some $n \geq 1$, i.e., $f(n) = 3^n - 1$. Then:

$$\begin{aligned} f(n+1) &= 3f(n) + 2 \\ &= 3(3^n - 1) + 2 \\ &= 3^{n+1} - 3 + 2 \\ &= 3^{n+1} - 1. \end{aligned}$$

Thus, the claim holds for all $n \geq 1$ by induction.

Teacher Bob has 10 students and 20 candies.

Each student assigns a distinct value to each candy, summing to 3230.

- › Bob sees all students' values and selects an ordering.
- › Students pick candies in **two rounds** based on Bob's order.
- › On each turn, a student picks their highest-value available candy.
- › Each student gets **2 candies**, with a **final value** equal to their sum.

Prove that Bob can always choose an ordering such that the sum of all 10 students' final values is at least 3230. $\square$

Solution

If the ordering is chosen **uniformly at random**, the expected total final value is at least 3230, implying that such an ordering must exist.

Expected Value Calculation

Fix a student with values for the candies $a_1 > a_2 > \dots > a_{20}$, where:

$$a_1 + a_2 + \dots + a_{20} = 3230.$$

- › In position j , the student picks at least a_j first.
- › In position $10 + j$, she picks at least a_{10+j} .
- › Her final value is at least $a_j + a_{10+j}$.

Since each position is equally likely ($1/10$), her expected final value is *at least*:

$$\frac{1}{10} \sum_{j=1}^{10} (a_j + a_{10+j}) = \frac{1}{10} \cdot 3230 = \frac{3230}{10}.$$

By **linearity of expectation**, the total expected value is *at least*: $10 \cdot \frac{3230}{10} = 3230$.

Attendance Taking

Charlie has **100 coins**, knowing that **4 are fake** but not which ones.

- › **All real coins** have the same weight.
- › **All fake coins** have the same weight, but are **lighter** than real coins.
- › Charlie does not know these weights.

Charlie's Balance

He can compare two **disjoint** sets of coins A and B , determining:

- 1 $A > B$: A is heavier than B
- 2 $A < B$: A is lighter than B .
- 3 $A = B$: A and B weigh equally.

Determine, with proof, a small number k such that by using at most k weighings, Charlie can always point to one coin and say with certainty that this coin is real. $\square$

Suboptimal Solutions

- $k = 4$: Take any 5 coins. At least one of them must be real. We can find a real coin using at most 4 weighings. The idea is similar to **finding the maximum element** in a length- n array using $n - 1$ comparisons.
- $k = 3$: Take any 8 coins. At least one of them must be real. We can find a real coin using at most 3 weighings. The idea is to leverage **divide-and-conquer**.

So far, we haven't used the full 100 coins or the fact that there are exactly 4 fake coins. The optimal solution achieves $k = 2$, and it will need to utilize this information, carefully...

Solution

Charlie can determine the fake coins in at most $k = 2$ **weighings**.

Step 1: Initial Weighing

Divide 100 coins into $A = 33$, $B = 33$, $C = 34$.

- › Weigh A vs. B (If **unequal** - at most **1 fake coin** is in heavier set)
 - ›› Remove a coin from the heavier set (1 coins),
 - ›› Weigh the rest in two equal sets (16 coins).
- › Weigh A vs. B (If **equal** - C has **0, 2, or 4** fake coins)
 - ›› Weigh $B \cup \{x\}$ vs. C for some $x \in A$.

Step 2.Neq: Second Weighing (If first weighing unequal, say¹ $A > B$)

Set A is split 3 ways A_1, A_2, A_3 with respective coin sizes 16, 16, 1.

- › Weigh A_1 vs. A_2 (If **unequal**)
 - ›› Removed coin in A_3 is real.
- › Weigh A_1 vs. A_2 (If **equal**)
 - ›› Coins in both A_1 and A_2 are real.

¹Note that $B > A$ is symmetric.

Step 2.Eq: Second Weighing (If first weighing equal)

Create set $B' = B \cup \{x\}$, by adding $x \in A$ to B .

- › Weigh B' vs. C (If $B' > C$)
 - ›› Added coin x is real.
- › Weigh B' vs. C (If $B' = C$)
 - ›› Coins in $A \setminus \{x\}$ are real.
- › Weigh B' vs. C (If $B' < C$)
 - ›› Coins in C are real.

It may be clearer to see an illustration of the decision tree of the 2 weighings with the possible configurations.

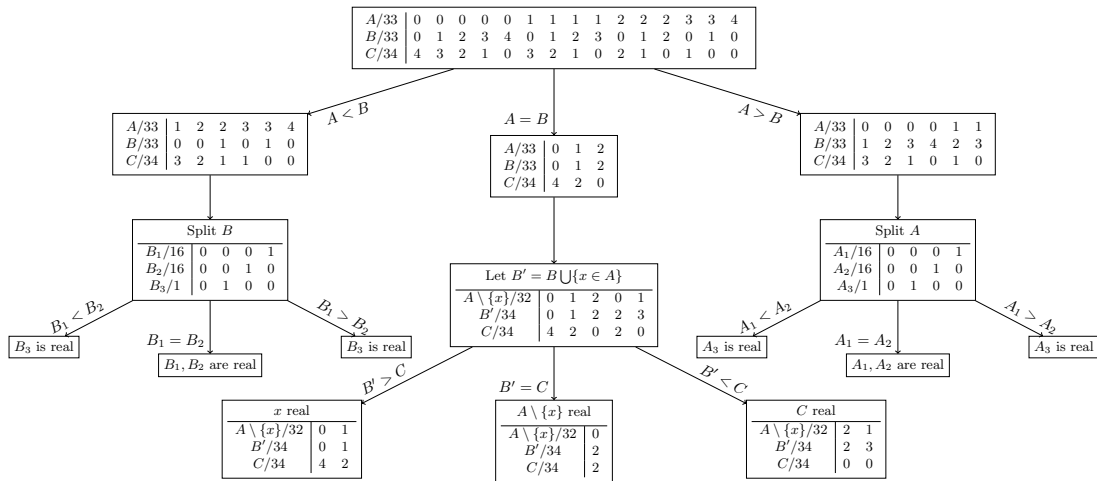


Figure 1: Configurations of fake coins across sets A , B , and C , where each column represents a unique combination. The table specifies the number of fake coins in each set (e.g., $A/33$ indicates that set A has 33 coins, with the corresponding cell showing the number of fake coins in A).

- Record class part



Figure 2: Feedback form

Thank you!

QnA